Please check the examination details below before entering your candidate information

Candidate surname	Other names

Centre Number | Candidate Number

Pearson Edexcel Level 3 GCE

Friday 16 May 2025

Afternoon

Paper reference **8FM0/25**

Further Mathematics

Advanced Subsidiary
Further Mathematics options
25: Further Mechanics 1
(Part of options C, E, H and J)

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Unless otherwise indicated, whenever a value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 4 questions.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ▶

1. A car of mass 1000 kg moves along a straight horizontal road at a constant speed U $\mathrm{m\,s^{-1}}$. The engine of the car is working at a rate of 20 kW.

The total resistance to the motion of the car is modelled as a constant force of magnitude 1600 N.

Using the model,

(a) find the value of U.

(3)

Later on, the car moves down a straight road which is inclined to the horizontal at an angle α, where $\sin \alpha = \frac{1}{49}$

The total resistance to the motion of the car is again modelled as a constant force of magnitude 1600 N.

At the instant when the engine of the car is working at a rate of 20 kW and the speed of the car is 8 m $\mathrm{s^{-1}}$, the acceleration of the car is a m $\mathrm{s^{-2}}$

Using the model,

(b) find the value of a.

(4)

(c) State one improvement to the model that would make it more realistic.

(1)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

Question 1 continued

Question 1 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 1 continued

(Total for Question 1 is 8 marks)

2. A particle Q of mass $3m$ is at rest on a smooth horizontal plane. A particle P of mass m is moving along the plane when it collides directly with Q.

The speed of P immediately before the collision is u.

The direction of motion of P is reversed by the collision.

The coefficient of restitution between P and Q is e.

(a) Show that the speed of P immediately after the collision is $\dfrac{u(3e-1)}{4}$

(6)

(b) State the full range of possible values of e.

(1)

Given that $e = \dfrac{1}{2}$

(c) find, in terms of m and u, the magnitude of the impulse exerted by P on Q in the collision.

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 2 continued

(Total for Question 2 is 10 marks)

Turn over

3. A plane is inclined to the horizontal at an angle α, where $\tan \alpha = \frac{4}{3}$. A small block of mass m is held at a point A on the plane and released from rest.

Initially the block is modelled as a particle, air resistance is modelled as being negligible and the plane is modelled as being **smooth**.

(a) Using the model and the principle of conservation of mechanical energy, find the distance of the block from A at the instant when the speed of the block is $7\,\text{m}\,\text{s}^{-1}$

(3)

In a refined model, the block is again modelled as a particle and air resistance is modelled as being negligible, but the plane is modelled as being **rough** with the coefficient of friction between the block and the plane being $\frac{2}{3}$

Using the refined model,

(b) find, in terms of m and g, the magnitude of the frictional force acting on the block as it slides down the plane,

(3)

(c) find, using the work-energy principle, the distance of the block from A at the instant when the speed of the block is $7\,\text{m}\,\text{s}^{-1}$

(4)

Question 3 continued

Question 3 continued

Question 3 continued

(Total for Question 3 is 10 marks)

4.

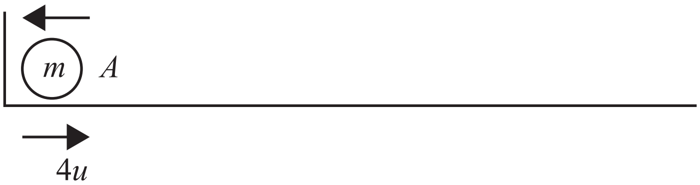

Figure 1

A particle A of mass m is moving on a smooth horizontal plane when it collides directly with a fixed vertical wall which is perpendicular to the direction of motion of A. Immediately **after** the collision, the speed of A is $4u$, directly away from the wall, as shown in Figure 1.

The coefficient of restitution between A and the wall is e.

Given that the magnitude of the impulse exerted on A by the wall in the collision is $9mu$,

(a) find the value of e.

(5)

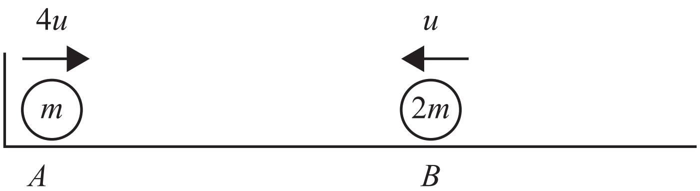

Figure 2

At the instant when A rebounds from the wall, a particle B of mass $2m$ is projected along the plane towards A, as shown in Figure 2.

The particles are moving in opposite directions along the same straight line when they collide directly.
Immediately **before** the collision, the speed of A is $4u$ and the speed of B is u.
Immediately **after** the collision, the total kinetic energy of the two particles is mu^2

(b) Determine, showing your method clearly, whether there are any further collisions between A and the wall.

(7)

Question 4 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 4 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 4 continued

Question 4 continued

(Total for Question 4 is 12 marks)

TOTAL FOR FURTHER MECHANICS 1 IS 40 MARKS

Please check the examination details below before entering your candidate information

Candidate surname	Other names

Centre Number | Candidate Number

Pearson Edexcel Level 3 GCE

Friday 17 May 2024

Afternoon

Paper reference **8FM0/25**

Further Mathematics

Advanced Subsidiary
Further Mathematics options
25: Further Mechanics 1
(Part of options C, E, H and J)

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Unless otherwise indicated, whenever a value of g is required, take $g = 9.8\,\text{m}\,\text{s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 4 questions.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over

Pearson

1. A particle A has mass $2m$ and a particle B has mass $3m$. The particles are moving in opposite directions along the same straight line and collide directly.

Immediately **before** the collision, the speed of A is $2u$ and the speed of B is u.
Immediately **after** the collision, the speed of A is $0.5u$ and the speed of B is w.

Given that the direction of motion of each particle is reversed by the collision,

(a) find w in terms of u

(3)

(b) find the coefficient of restitution between the particles,

(3)

(c) find, in terms of m and u, the magnitude of the impulse received by A in the collision.

(3)

Question 1 continued

(Total for Question 1 is 9 marks)

2. A lorry has mass 5000 kg.

In all circumstances, when the speed of the lorry is $v\,\text{m s}^{-1}$, the resistance to motion of the lorry from non-gravitational forces is modelled as having magnitude $490v$ newtons.

The lorry moves along a straight horizontal road at $12\,\text{m s}^{-1}$, with its engine working at a constant rate of 84 kW.

Using the model,

(a) find the acceleration of the lorry.

(4)

Another straight road is inclined to the horizontal at an angle α where $\sin\alpha = \dfrac{1}{14}$

With its engine again working at a constant rate of 84 kW, the lorry can maintain a constant speed of $V\,\text{m s}^{-1}$ up the road.

Using the model,

(b) find the value of V.

(4)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 2 continued

(Total for Question 2 is 8 marks)

3.

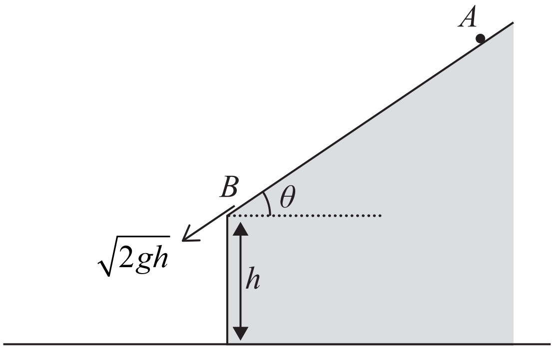

Figure 1

Figure 1 shows part of the end elevation of a building which sits on horizontal ground. The side of the building is vertical and has height h.

A small stone of mass m is at rest on the roof of the building at the point A. The stone slides from rest down a line of greatest slope of the roof and reaches the edge B of the roof with speed $\sqrt{2gh}$

The stone then moves under gravity before hitting the ground with speed W.

In a model of the motion of the stone **from *B* to the ground**

- the stone is modelled as a particle
- air resistance is ignored

Using the principle of conservation of mechanical energy and the model,

(a) find W in terms of g and h.

(4)

In a model of the motion of the stone **from *A* to *B***

- the stone is modelled as a particle of mass m
- air resistance is ignored
- the roof of the building is modelled as a rough plane inclined to the horizontal at an angle θ, where $\tan\theta = \frac{3}{4}$
- the coefficient of friction between the stone and the roof is $\frac{1}{3}$
- $AB = d$

Using this model,

(b) find, in terms of m and g, the magnitude of the frictional force acting on the stone as it slides down the roof,

(3)

(c) use the work–energy principle to find d in terms of h.

(5)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

Question 3 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 3 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 3 continued

(Total for Question 3 is 12 marks)

4.

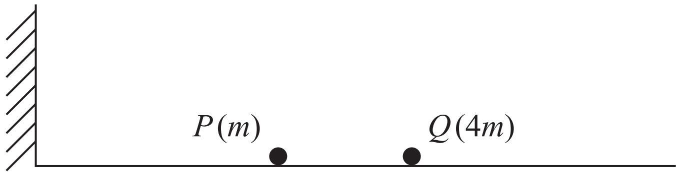

Figure 2

A particle P of mass m and a particle Q of mass $4m$ are at rest on a smooth horizontal plane, as shown in Figure 2.

Particle P is projected with speed u along the plane towards Q and the particles collide.

The coefficient of restitution between the particles is e, where $e > \frac{1}{4}$

As a result of the collision, the direction of motion of P is reversed and P has speed $\frac{u}{5}(4e - 1)$.

(a) Find, in terms of u and e, the speed of Q after the collision.

(3)

After the collision, P goes on to hit a vertical wall which is fixed at right angles to the direction of motion of P.

The coefficient of restitution between P and the wall is f, where $f > 0$

Given that $e = \frac{3}{4}$

(b) find, in terms of m, u and f, the kinetic energy lost by P as a result of its impact with the wall. Give your answer in its simplest form.

(4)

After its impact with the wall, P goes on to collide with Q again.

(c) Find the complete range of possible values of f.

(4)

Question 4 continued

Question 4 continued

(Total for Question 4 is 11 marks)

TOTAL FOR FURTHER MECHANICS 1 IS 40 MARKS

Please check the examination details below before entering your candidate information

Candidate surname	Other names

Centre Number | Candidate Number

Pearson Edexcel Level 3 GCE

Friday 19 May 2023

Afternoon

Paper reference **8FM0/25**

Further Mathematics

Advanced Subsidiary
Further Mathematics options
25: Further Mechanics 1
(Part of options C, E, H and J)

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Unless otherwise indicated, whenever a value of g is required, take $g = 9.8\,\text{m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 4 questions.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over

P72811A

N:1/1/1/1/

P 7 2 8 1 1 A 0 1 1 6

Pearson

1. Two particles, P and Q, of masses $3m$ and $2m$ respectively, are moving on a smooth horizontal plane. They are moving in opposite directions along the same straight line when they collide directly.

Immediately before the collision, P is moving with speed $2u$.

The magnitude of the impulse exerted on P by Q in the collision is $\dfrac{9mu}{2}$

(a) Find the speed of P immediately after the collision.

(3)

The coefficient of restitution between P and Q is e.

Given that the speed of Q immediately before the collision is u,

(b) find the value of e.

(5)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

Question 1 continued

(Total for Question 1 is 8 marks)

P 7 2 8 1 1 A 0 3 1 6

2. A racing car of mass 750 kg is moving along a straight horizontal road at a constant speed of U **km h^{-1}**. The engine of the racing car is working at a constant rate of 60 kW.

The resistance to the motion of the racing car is modelled as a force of magnitude $37.5v$ N, where v m s^{-1} is the speed of the racing car.

Using the model,

(a) find the value of U

(4)

Later on, the racing car is accelerating up a straight road which is inclined to the horizontal at an angle α, where $\sin\alpha = \frac{5}{49}$. The engine of the racing car is working at a constant rate of 60 kW.

The total resistance to the motion of the racing car from non-gravitational forces is modelled as a force of magnitude $37.5v$ N, where v m s^{-1} is the speed of the racing car. At the instant when the acceleration of the racing car is 2 m s^{-2}, the speed of the racing car is V m s^{-1}

Using the model,

(b) find the value of V

(4)

Question 2 continued

Question 2 continued

Question 2 continued

(Total for Question 2 is 8 marks)

3. A stone of mass 0.5 kg is projected vertically upwards with a speed $U\,\text{m s}^{-1}$ from a point A. The point A is 2.5 m above horizontal ground.

The speed of the stone as it hits the ground is $25\,\text{m s}^{-1}$

The motion of the stone from the instant it is projected from A until the instant it hits the ground is modelled as that of a particle moving freely under gravity.

(a) Use the model and the principle of conservation of mechanical energy to find the value of U.

(4)

In reality, the stone will be subject to air resistance as it moves from A to the ground.

(b) State how this would affect your answer to part (a).

(1)

The ground is soft and the stone sinks a vertical distance d cm into the ground. The resistive force exerted on the stone by the ground is modelled as a constant force of magnitude 2000 N and the stone is modelled as a particle.

(c) Use the model and the work-energy principle to find the value of d, giving your answer to 3 significant figures.

(5)

Question 3 continued

Question 3 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 3 continued

(Total for Question 3 is 10 marks)

4.

Figure 1

Three particles, P, Q and R, lie at rest on a smooth horizontal plane. The particles are in a straight line with Q between P and R, as shown in Figure 1.

Particle P is projected **towards** Q with speed u. At the same time, R is projected with speed $\frac{1}{2}u$ **away from** Q, in the direction QR.

Particle P has mass m and particle Q has mass $2m$.

The coefficient of restitution between P and Q is e.

(a) Show that the speed of Q immediately after the collision between P and Q is

$\dfrac{u(1+e)}{3}$ **(6)**

It is given that $e > \dfrac{1}{2}$

(b) Determine whether there is a collision between Q and R. **(2)**

(c) Determine the direction of motion of P immediately after the collision between P and Q. **(2)**

(d) Find, in terms of m, u and e, the total kinetic energy lost in the collision between P and Q, simplifying your answer. **(3)**

(e) Explain how using $e = 1$ could be used to check your answer to part (d). **(1)**

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 4 continued

Question 4 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 4 continued

Question 4 continued

(Total for Question 4 is 14 marks)

TOTAL FOR FURTHER MECHANICS 1 IS 40 MARKS

Please check the examination details below before entering your candidate information

Candidate surname	Other names

Centre Number | Candidate Number

Pearson Edexcel Level 3 GCE

Paper reference **8FM0/25**

Further Mathematics

Advanced Subsidiary
Further Mathematics options
25: Further Mechanics 1
(Part of options C, E, H and J)

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Unless otherwise indicated, whenever a value of g is required, take $g = 9.8\,\text{m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 4 questions.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ▶

1. A car of mass 1200 kg moves up a straight road that is inclined to the horizontal at an angle α, where $\sin\alpha = \dfrac{1}{15}$

The total resistance to the motion of the car from non-gravitational forces is modelled as a constant force of magnitude R newtons.

At the instant when the engine of the car is working at a rate of 32 kW and the speed of the car is $20\,\mathrm{m\,s^{-1}}$, the acceleration of the car is $0.5\,\mathrm{m\,s^{-2}}$

Find the value of R

(5)

Question 1 continued

(Total for Question 1 is 5 marks)

2. Two particles, A and B, have masses m and $3m$ respectively. The particles are moving in opposite directions along the same straight line on a smooth horizontal plane when they collide directly.

Immediately before they collide, A is moving with speed $2u$ and B is moving with speed u.

The direction of motion of each particle is reversed by the collision.

In the collision, the magnitude of the impulse exerted on A by B is $\dfrac{9mu}{2}$

(a) Find the value of the coefficient of restitution between A and B.
(7)

(b) Hence, write down the total loss in kinetic energy due to the collision, giving a reason for your answer.
(1)

Question 2 continued

Question 2 continued

Question 2 continued

(Total for Question 2 is 8 marks)

3. A plane is inclined to the horizontal at an angle α, where $\tan\alpha = \frac{3}{4}$

A particle P is held at rest at a point A on the plane.

The particle P is then projected with speed $25\,\mathrm{m\,s^{-1}}$ from A, up a line of greatest slope of the plane.

In an initial model, the plane is modelled as being smooth and air resistance is modelled as being negligible.

Using this model and the principle of conservation of mechanical energy,

(a) find the speed of P at the instant when it has travelled a distance $\frac{25}{6}$ m up the plane from A.

(4)

In a refined model, the plane is now modelled as being rough, with the coefficient of friction between P and the plane being $\frac{3}{5}$

Air resistance is still modelled as being negligible.

Using this refined model and the work-energy principle,

(b) find the speed of P at the instant when it has travelled a distance $\frac{25}{6}$ m up the plane from A.

(8)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

Question 3 continued

Question 3 continued

Question 3 continued

(Total for Question 3 is 12 marks)

4. A particle P of mass $2m$ kg is moving with speed $2u\,\mathrm{m\,s^{-1}}$ on a smooth horizontal plane. Particle P collides with a particle Q of mass $3m$ kg which is at rest on the plane. The coefficient of restitution between P and Q is e. Immediately after the collision the speed of Q is $v\,\mathrm{m\,s^{-1}}$

(a) Show that $v = \dfrac{4u(1+e)}{5}$

(6)

(b) Show that $\dfrac{4u}{5} \leqslant v \leqslant \dfrac{8u}{5}$

(2)

Given that the direction of motion of P is reversed by the collision,

(c) find, in terms of u and e, the speed of P immediately after the collision.

(2)

After the collision, Q hits a wall, that is fixed at right angles to the direction of motion of Q, and rebounds.

The coefficient of restitution between Q and the wall is $\dfrac{1}{6}$

Given that P and Q collide again,

(d) find the full range of possible values of e.

(5)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

Question 4 continued

Question 4 continued

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

Question 4 continued

Question 4 continued

(Total for Question 4 is 15 marks)

TOTAL FOR FURTHER MECHANICS 1 IS 40 MARKS

Please check the examination details below before entering your candidate information

Candidate surname	Other names

Pearson Edexcel Level 3 GCE

Centre Number | Candidate Number

Paper reference **8FM0/25**

Further Mathematics

Advanced Subsidiary
Further Mathematics options
25: Further Mechanics 1
(Part of options C, E, H and J)

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by the regulations of the Joint Council for Qualifications. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Unless otherwise indicated, whenever a value of g is required, take $g = 9.8\,\text{m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 4 questions.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.
- Good luck with your examination.

Turn over

Pearson

1.

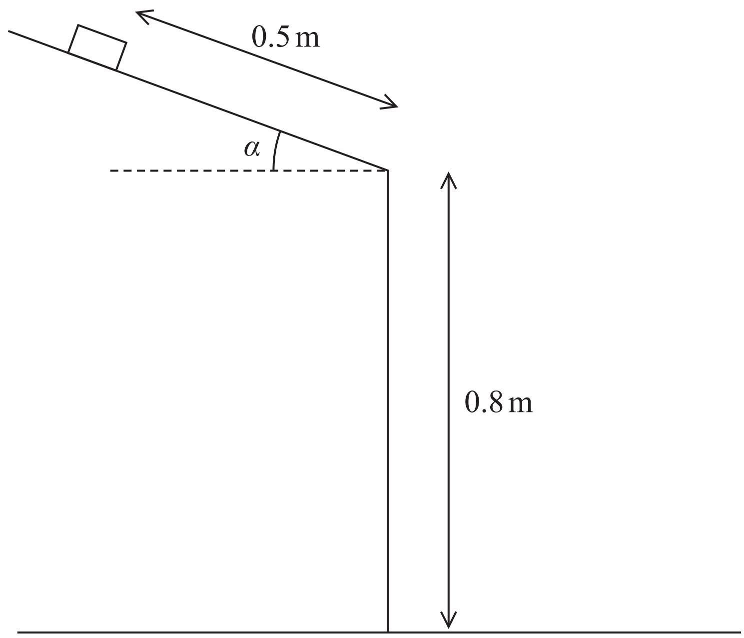

Figure 1

A small book of mass m is held on a rough straight desk lid which is inclined at an angle α to the horizontal, where $\tan\alpha = \frac{3}{4}$. The book is released from rest at a distance of 0.5 m from the edge of the desk lid, as shown in Figure 1. The book slides down the desk lid and then hits the floor that is 0.8 m below the edge of the desk lid. The coefficient of friction between the book and the desk lid is 0.4

The book is modelled as a particle which, after leaving the desk lid, is assumed to move freely under gravity.

(a) Find, in terms of m and g, the magnitude of the normal reaction on the book as it slides down the desk lid.

(2)

(b) Use the work-energy principle to find the speed of the book as it hits the floor.

(5)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

Question 1 continued

Question 1 continued

Question 1 continued

(Total for Question 1 is 7 marks)

2.

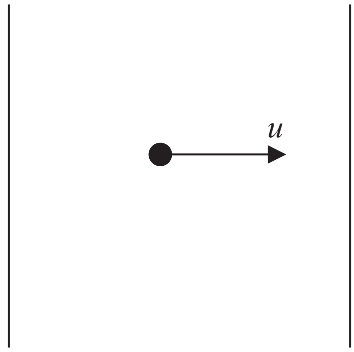

Figure 2

A particle of mass em is at rest on a smooth horizontal plane between two smooth fixed parallel vertical walls, as shown in the plan view in Figure 2. The particle is projected along the plane with speed u towards one of the walls and strikes the wall at right angles. The coefficient of restitution between the particle and each wall is e and air resistance is modelled as being negligible.

Using the model,

(a) find, in terms of m, u and e, an expression for the total loss in the kinetic energy of the particle as a result of the first two impacts.

(3)

Given that e can vary such that $0 < e < 1$ and using the model,

(b) find the value of e for which the total loss in the kinetic energy of the particle as a result of the first two impacts is a maximum,

(4)

(c) describe the subsequent motion of the particle.

(2)

Question 2 continued

Question 2 continued

Question 2 continued

(Total for Question 2 is 9 marks)

3. The total mass of a cyclist and his bicycle is 100 kg.

In all circumstances, the magnitude of the resistance to the motion of the cyclist from non-gravitational forces is modelled as being kv^2 N, where v m s^{-1} is the speed of the cyclist.

The cyclist can freewheel, without pedalling, down a slope that is inclined to the horizontal at an angle α, where $\sin\alpha = \frac{1}{35}$, at a constant speed of V m s^{-1}

When he is pedalling up a slope that is inclined to the horizontal at an angle β, where $\sin\beta = \frac{1}{70}$, and he is moving at the same constant speed V m s^{-1}, he is working at a constant rate of P watts.

(a) Find P in terms of V.

(7)

If he pedals and works at a rate of $35V$ watts on a horizontal road, he moves at a constant speed of U m s^{-1}

(b) Find U in terms of V.

(4)

Question 3 continued

Question 3 continued

Question 3 continued

(Total for Question 3 is 11 marks)

4. Two particles, P and Q, have masses m and em respectively. The particles are moving on a smooth horizontal plane in the same direction along the same straight line when they collide directly. The coefficient of restitution between P and Q is e, where $0 < e < 1$

Immediately before the collision the speed of P is u and the speed of Q is eu.

(a) Show that the speed of Q immediately after the collision is u.

(6)

(b) Show that the direction of motion of P is unchanged by the collision.

(3)

The magnitude of the impulse on Q in the collision is $\dfrac{2}{9}mu$

(c) Find the possible values of e.

(4)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 4 continued

Question 4 continued

(Total for Question 4 is 13 marks)

TOTAL FOR FURTHER MECHANICS 1 IS 40 MARKS

Please check the examination details below before entering your candidate information

Candidate surname	Other names

Pearson Edexcel Level 3 GCE

Centre Number

Candidate Number

Thursday 08 October 2020

Afternoon

Paper Reference **8FM0/25**

Further Mathematics

Advanced Subsidiary

Further Mathematics options

25: Further Mechanics 1

(Part of options C, E, H and J)

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Unless otherwise indicated, whenever a value of g is required, take $g = 9.8\,\text{m}\,\text{s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 4 questions.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ▶

1/1/1/1/1/
P 6 1 8 6 8 A 0 1 1 6

Pearson

1. Two particles P and Q have masses m and $4m$ respectively. The particles are at rest on a smooth horizontal plane. Particle P is given a horizontal impulse, of magnitude I, in the direction PQ. Particle P then collides directly with Q. Immediately after this collision, P is at rest and Q has speed w. The coefficient of restitution between the particles is e.

(a) Find I in terms of m and w.

(2)

(b) Show that $e = \dfrac{1}{4}$

(1)

(c) Find, in terms of m and w, the total kinetic energy lost in the collision between P and Q.

(2)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 1 continued

(Total for Question 1 is 5 marks)

2. A car of mass 1000 kg moves along a straight horizontal road.

In all circumstances, when the speed of the car is $v\,\mathrm{m\,s^{-1}}$, the resistance to the motion of the car is modelled as a force of magnitude cv^2 N, where c is a constant.

The maximum power that can be developed by the engine of the car is 50 kW.

At the instant when the speed of the car is $72\,\mathrm{km\,h^{-1}}$ and the engine is working at its maximum power, the acceleration of the car is $2.25\,\mathrm{m\,s^{-2}}$

(a) Convert $72\,\mathrm{km\,h^{-1}}$ into $\mathrm{m\,s^{-1}}$

(1)

(b) Find the acceleration of the car at the instant when the speed of the car is $144\,\mathrm{km\,h^{-1}}$ and the engine is working at its maximum power.

(7)

The maximum speed of the car when the engine is working at its maximum power is $V\,\mathrm{km\,h^{-1}}$.

(c) Find, to the nearest whole number, the value of V.

(4)

Question 2 continued

Question 2 continued

Question 2 continued

(Total for Question 2 is 12 marks)

3. Three particles A, B and C are at rest on a smooth horizontal plane. The particles lie along a straight line with B between A and C.

Particle B has mass $4m$ and particle C has mass km, where k is a positive constant. Particle B is projected with speed u along the plane towards C and they collide directly.

The coefficient of restitution between B and C is $\frac{1}{4}$

(a) Find the range of values of k for which there would be no further collisions.

(8)

The magnitude of the impulse on B in the collision between B and C is $3mu$

(b) Find the value of k.

(4)

Question 3 continued

Question 3 continued

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

Question 3 continued

(Total for Question 3 is 12 marks)

Turn over

4. A small ball, of mass m, is thrown vertically upwards with speed $\sqrt{8gH}$ from a point O on a smooth horizontal floor. The ball moves towards a smooth horizontal ceiling that is a vertical distance H above O. The coefficient of restitution between the ball and the ceiling is $\frac{1}{2}$

In a model of the motion of the ball, it is assumed that the ball, as it moves up or down, is subject to air resistance of constant magnitude $\frac{1}{2}mg$.

Using this model,

(a) use the work-energy principle to find, in terms of g and H, the speed of the ball immediately before it strikes the ceiling,

(5)

(b) find, in terms of g and H, the speed of the ball immediately before it strikes the floor at O for the first time.

(5)

In a simplified model of the motion of the ball, it is assumed that the ball, as it moves up or down, is subject to no air resistance.

Using this simplified model,

(c) explain, without any detailed calculation, why the speed of the ball, immediately before it strikes the floor at O for the first time, would still be less than $\sqrt{8gH}$

(1)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

Question 4 continued

Question 4 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 4 continued

Question 4 continued

(Total for Question 4 is 11 marks)

TOTAL FOR FURTHER MECHANICS 1 IS 40 MARKS

Please check the examination details below before entering your candidate information

Candidate surname	Other names

Pearson Edexcel Level 3 GCE

Centre Number | Candidate Number

Thursday 16 May 2019

Afternoon | Paper Reference **8FM0-25**

Further Mathematics

Advanced Subsidiary
Further Mathematics options
25: Further Mechanics 1
(Part of options C, E, H and J)

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Unless otherwise indicated, whenever a value of g is required, take $g = 9.8\text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 4 questions.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ▶

1/1/1/1/1/1/1

Pearson

1. A lorry of mass 16 000 kg moves along a straight horizontal road.

The lorry moves at a constant speed of 25 $\mathrm{m\,s^{-1}}$

In an initial model for the motion of the lorry, the resistance to the motion of the lorry is modelled as having constant magnitude 16 000 N.

(a) Show that the engine of the lorry is working at a rate of 400 kW.

(4)

The model for the motion of the lorry along the same road is now refined so that when the speed of the lorry along the same road is V $\mathrm{m\,s^{-1}}$, the resistance to the motion of the lorry is modelled as having magnitude $640V$ newtons.

Assuming that the engine of the lorry is working at the same rate of 400 kW

(b) use the refined model to find the speed of the lorry when it is accelerating at 2.1 $\mathrm{m\,s^{-2}}$

(6)

Question 1 continued

Question 1 continued

Question 1 continued

(Total for Question 1 is 10 marks)

2. Two particles, A and B, of masses $2m$ and $3m$ respectively, are moving on a smooth horizontal plane. The particles are moving in opposite directions towards each other along the same straight line when they collide directly. Immediately before the collision the speed of A is $2u$ and the speed of B is u. In the collision the impulse of A on B has magnitude $5mu$.

(a) Find the coefficient of restitution between A and B.

(9)

(b) Find the total loss in kinetic energy due to the collision.

(4)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

Question 2 continued

Question 2 continued

Question 2 continued

(Total for Question 2 is 13 marks)

3. A particle, P, of mass m kg is projected with speed 5 m s^{-1} down a line of greatest slope of a rough plane. The plane is inclined to the horizontal at an angle α, where $\sin\alpha = \dfrac{3}{5}$

The total resistance to the motion of P is a force of magnitude $\dfrac{1}{5}\,mg$

Use the work-energy principle to find the speed of P at the instant when it has moved a distance 8 m down the plane from the point of projection.

(7)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 3 continued

(Total for Question 3 is 7 marks)

4. Three particles, P, Q and R, are at rest on a smooth horizontal plane. The particles lie along a straight line with Q between P and R. The particles Q and R have masses m and km respectively, where k is a constant.

Particle Q is projected towards R with speed u and the particles collide directly.

The coefficient of restitution between each pair of particles is e.

(a) Find, in terms of e, the range of values of k for which there is a second collision.

(9)

Given that the mass of P is km and that there is a second collision,

(b) write down, in terms of u, k and e, the speed of Q after this second collision.

(1)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

Question 4 continued

Question 4 continued

Question 4 continued

Question 4 continued

(Total for Question 4 is 10 marks)

TOTAL FOR FURTHER MECHANICS 1 IS 40 MARKS

Write your name here

Surname	Other names

Pearson Edexcel
Level 3 GCE

Centre Number	Candidate Number

Further Mathematics
Advanced Subsidiary
Further Mathematics options
25: Further Mechanics 1
(Part of options C, E, H and J)

Thursday 17 May 2018 – Afternoon

Paper Reference
8FM0-25

You must have:
Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator allowed by the regulations of the Joint Council for Qualifications. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions
- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information
- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 4 questions.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice
- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ▶

Pearson

Whenever a numerical value of g is required, take $g = 9.8\,\text{m}\,\text{s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Answer ALL questions. Write your answers in the spaces provided.

1. A small ball of mass 0.3 kg is released from rest from a point 3.6 m above horizontal ground. The ball falls freely under gravity, hits the ground and rebounds vertically upwards.

In the first impact with the ground, the ball receives an impulse of magnitude 4.2 N s. The ball is modelled as a particle.

(a) Find the speed of the ball immediately after it first hits the ground.

(5)

(b) Find the kinetic energy lost by the ball as a result of the impact with the ground.

(3)

Question 1 continued

(Total for Question 1 is 8 marks)

2.

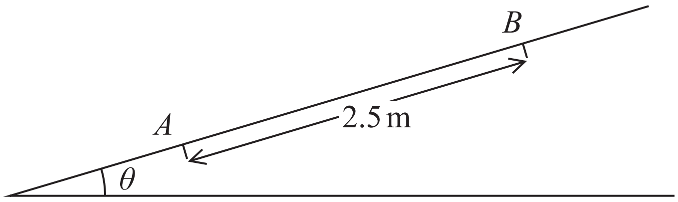

Figure 1

Figure 1 shows a ramp inclined at an angle θ to the horizontal, where $\sin\theta = \frac{2}{7}$

A parcel of mass 4 kg is projected, with speed $5\,\mathrm{m\,s^{-1}}$, from a point A on the ramp.
The parcel moves up a line of greatest slope of the ramp and first comes to instantaneous rest at the point B, where $AB = 2.5$ m.
The parcel is modelled as a particle.

The total resistance to the motion of the parcel from non-gravitational forces is modelled as a constant force of magnitude R newtons.

(a) Use the work-energy principle to show that $R = 8.8$

(4)

After coming to instantaneous rest at B, the parcel slides back down the ramp. The total resistance to the motion of the particle is modelled as a constant force of magnitude 8.8 N.

(b) Find the speed of the parcel at the instant it returns to A.

(3)

(c) Suggest two improvements that could be made to the model.

(2)

Question 2 continued

Question 2 continued

Question 2 continued

(Total for Question 2 is 9 marks)

3. A van of mass 750 kg is moving along a straight horizontal road. At the instant when the van is moving at $v\,\text{m}\,\text{s}^{-1}$, the resistance to the motion of the van is modelled as a force of magnitude λv N, where λ is a constant.

The engine of the van is working at a constant rate of 18 kW.
At the instant when $v = 15$, the acceleration of the van is $0.6\,\text{m}\,\text{s}^{-2}$

(a) Show that $\lambda = 50$

(4)

The van now moves up a straight road inclined at an angle to the horizontal, where
$\sin\alpha = \dfrac{1}{15}$

At the instant when the van is moving at $v\,\text{m}\,\text{s}^{-1}$, the resistance to the motion of the van from non-gravitational forces is modelled as a force of magnitude $50v$ N.
When the engine of the van is working at a constant rate of 12 kW, the van is moving at a constant speed $V\,\text{m}\,\text{s}^{-1}$

(b) Find the value of V.

(5)

Question 3 continued

Question 3 continued

Question 3 continued

(Total for Question 3 is 9 marks)

4. A particle P of mass $3m$ is moving in a straight line on a smooth horizontal floor. A particle Q of mass $5m$ is moving in the opposite direction to P along the same straight line.

The particles collide directly.

Immediately before the collision, the speed of P is $2u$ and the speed of Q is u.
The coefficient of restitution between P and Q is e.

(a) Show that the speed of Q immediately after the collision is $\frac{u}{8}(9e + 1)$

(6)

(b) Find the range of values of e for which the direction of motion of P is not changed as a result of the collision.

(2)

When P and Q collide they are at a distance d from a smooth fixed vertical wall, which is perpendicular to their direction of motion. After the collision with P, particle Q collides directly with the wall and rebounds so that there is a second collision between P and Q. This second collision takes place at a distance x from the wall.

Given that $e = \frac{1}{18}$ and the coefficient of restitution between Q and the wall is $\frac{1}{3}$

(c) find x in terms of d.

(6)

Question 4 continued

Question 4 continued

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

P 6 0 2 0 6 A 0 1 4 1 6

Question 4 continued

Question 4 continued

(Total for Question 4 is 14 marks)

TOTAL FOR FURTHER MECHANICS 1 IS 40 MARKS

Write your name here

Surname	Other names

Pearson Edexcel Level 3 GCE

Centre Number | Candidate Number

Further Mathematics

Advanced Subsidiary
Further Mathematics options
Paper 2C: Further Pure Mathematics 1 and Further Mechanics 1

Sample Assessment Material for first teaching September 2017
Time: 1 hour 40 minutes

Paper Reference
8FM0/2C

You must have:
Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- There are **two** sections in this question paper. Answer **all** the questions in Section A and **all** the questions in Section B.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet ‘Mathematical Formulae and Statistical Tables’ is provided.
- There are 9 questions in this question paper. The total mark for this paper is 80.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ▶

S58540A

1/1/1/1/1/

S 5 8 5 4 0 A 0 1 2 2

Pearson

SECTION B

Answer ALL questions. Write your answers in the spaces provided.

Unless otherwise indicated, whenever a numerical value of g is required, take $g = 9.8\,\text{m}\,\text{s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

6. A small ball of mass 0.1 kg is dropped from a point which is 2.4 m above a horizontal floor. The ball falls freely under gravity, strikes the floor and bounces to a height of 0.6 m above the floor. The ball is modelled as a particle.

(a) Show that the coefficient of restitution between the ball and the floor is 0.5

(6)

(b) Find the height reached by the ball above the floor after it bounces on the floor for the second time.

(3)

(c) By considering your answer to (b), describe the subsequent motion of the ball.

(1)

Question 6 continued

(Total for Question 6 is 10 marks)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

7. A small stone of mass 0.5 kg is thrown vertically upwards from a point A with an initial speed of $25\,\mathrm{m\,s^{-1}}$. The stone first comes to instantaneous rest at the point B which is 20 m vertically above the point A. As the stone moves it is subject to air resistance. The stone is modelled as a particle.

(a) Find the energy lost due to air resistance by the stone, as it moves from A to B.

(3)

The air resistance is modelled as a constant force of magnitude R newtons.

(b) Find the value of R.

(2)

(c) State how the model for air resistance could be refined to make it more realistic.

(1)

Question 7 continued

(Total for Question 7 is 6 marks)

8. [*In this question use* $g = 10\,\mathrm{m\,s^{-2}}$]

A jogger of mass 60 kg runs along a straight horizontal road at a constant speed of $4\,\mathrm{m\,s^{-1}}$. The total resistance to the motion of the jogger is modelled as a constant force of magnitude 30 N.

(a) Find the rate at which the jogger is working.

(3)

The jogger now comes to a hill which is inclined to the horizontal at an angle α, where $\sin\alpha = \frac{1}{15}$. Because of the hill, the jogger reduces her speed to $3\,\mathrm{m\,s^{-1}}$ and maintains this constant speed as she runs up the hill. The total resistance to the motion of the jogger from non-gravitational forces continues to be modelled as a constant force of magnitude 30 N.

(b) Find the rate at which she has to work in order to run up the hill at $3\,\mathrm{m\,s^{-1}}$.

(5)

Question 8 continued

(Total for Question 8 is 8 marks)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

9. A particle P of mass $3m$ is moving in a straight line on a smooth horizontal table. A particle Q of mass m is moving in the opposite direction to P along the same straight line. The particles collide directly. Immediately before the collision the speed of P is u and the speed of Q is $2u$. The velocities of P and Q immediately after the collision, measured in the direction of motion of P before the collision, are v and w respectively. The coefficient of restitution between P and Q is e.

(a) Find an expression for v in terms of u and e.

(6)

Given that the direction of motion of P is changed by the collision,

(b) find the range of possible values of e.

(2)

(c) Show that $w = \frac{u}{4}(1 + 9e)$.

(2)

Following the collision with P, the particle Q then collides with and rebounds from a fixed vertical wall which is perpendicular to the direction of motion of Q. The coefficient of restitution between Q and the wall is f.

Given that $e = \frac{5}{9}$, and that P and Q collide again in the subsequent motion,

(d) find the range of possible values of f.

(6)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

Question 9 continued

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 9 continued

Question 9 continued

(Total for Question 9 is 16 marks)

TOTAL FOR SECTION B IS 40 MARKS
TOTAL FOR PAPER IS 80 MARKS

Write your name here

Surname	Other names

Pearson Edexcel Level 3 GCE

Centre Number	Candidate Number

Specimen Paper

Paper Reference **8FM0-25**

Further Mathematics
Advanced Subsidiary 25: Further Mechanics 1

You must have:
Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator allowed by the regulations of the Joint Council for Qualifications. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions
- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information
- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 4 questions.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*

Advice
- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ▶

Pearson

Unless otherwise indicated, whenever a value of g is required, take $g = 9.8\,\mathrm{m\,s^{-2}}$ and give your answer to either 2 significant figures or 3 significant figures.

Answer ALL questions. Write your answers in the spaces provided.

1. A ball is projected with speed $6\,\mathrm{m\,s^{-1}}$ up a line of greatest slope of an inclined plane.

The plane is inclined to the horizontal at an angle α, where $\sin\alpha = \frac{1}{7}$

The ball is modelled as a particle, the plane is modelled as being smooth and air resistance is modelled as being negligible. Using the conservation of energy principle, find the speed of the ball at the instant when it has travelled a distance of 5 m up the plane.

(5)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

Question 1 continued

(Total for Question 1 is 5 marks)

2. A particle P of mass m is moving in a straight line on a rough horizontal table. The particle collides with a fixed vertical wall. Immediately before P collides with the wall, P is moving with speed u in a direction that is perpendicular to the wall.

In the collision, P receives an impulse of magnitude $\frac{5mu}{3}$

After the collision, the total resistance to the motion of P is modelled as a constant force of magnitude $\frac{mg}{6}$

(a) Find, in terms of u and g, the distance of P from the wall when P comes to rest.

(7)

(b) State how the model for the total resistance could be refined to make it more realistic.

(1)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

S 6 0 9 8 0 A 0 4 1 2

Question 2 continued

(Total for Question 2 is 8 marks)

3. A car of mass 500 kg is moving at a speed of $v\,\mathrm{m\,s^{-1}}$. The total non-gravitational resistance to the motion of the car is modelled as having magnitude $(5v + C)$ newtons, where C is a constant.

The car moves up a straight road which is inclined to the horizontal at an angle θ, where $\sin\theta = \frac{1}{14}$. When the engine of the car is working at a constant rate of 25 kW, the car is moving up the road at a constant speed of $20\,\mathrm{m\,s^{-1}}$.

(a) Find the value of C.

(5)

With the engine of the car again working at 25 kW, the car now moves along a straight horizontal road at a constant speed of $U\,\mathrm{m\,s^{-1}}$.

(b) Find the value of U, giving your answer to 2 significant figures.

(6)

Question 3 continued

(Total for Question 3 is 11 marks)

4. A particle P of mass m is at rest on smooth horizontal ground between two fixed parallel vertical walls. Another particle Q of mass m is moving in a straight line along the ground between the walls in a direction which is perpendicular to the walls. Particle Q collides with particle P directly. The coefficient of restitution between the particles is e and the speed of Q immediately before the collision is u.

(a) Show that the speed of P immediately after the collision is $\frac{u}{2}(1 + e)$

(6)

(b) Find the speed of Q immediately after the collision.

(1)

Given that the total kinetic energy lost in the collision between the two particles is $\frac{3mu^2}{16}$

(c) find the value of e.

(5)

Suppose now that the coefficient of restitution between the particles is 1 and that the coefficient of restitution between each particle and each wall is 1

(d) By considering at least two collisions between the particles, describe in detail what happens in the subsequent motion, giving reasons for your answer.

(4)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

S60980A0812

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

Question 4 continued

Question 4 continued

Question 4 continued

Question 4 continued

(Total for Question 4 is 16 marks)

TOTAL FOR PAPER IS 40 MARKS